Phonetic Transcription

Phonetic Transcription

- Graphemes cannot represent speech sounds
  - cat vs. cite vs. cello
  - cat vs. kite
  - thin, graph, attend

Phonetic Transcription

- IPA (International Phonetic Alphabet)
- cat vs. cite vs. cello
  /kæt/ vs. /saɪt/ vs. /tʃeloʊ/
- cat vs. kite
  /kæt/ vs. /kaɪt/
- thin, graph, attend
  /θɪn/, /græf/, /ətend/

Phonemes

Phoneme

- The smallest distinctive unit of sound
- Can change the meaning of a word
- Abstract representations of sounds

Consonants

Phonetic symbol	Word positions		
	Initial	**Medial**	**Final**
Consonants			
p	*p*ack	su*p*er	ma*p*
b	*b*ed	ru*bb*er	ro*b*
t	*t*ea	a*tt*ack	grea*t*
d	*d*ate	a*d*ore	goo*d*
k	*c*atch	pi*ck*ing	loo*k*
g	*g*ate	do*gg*y	ba*g*
f	*f*at	co*ff*ee	loa*f*
v	*v*ery	mo*v*ing	do*v*e
θ	*th*in	ru*th*less	dea*th*
ð	*th*ey	mo*th*er	brea*th*e
s	*s*ad	si*s*ter	bu*s*
z	*z*oom	rai*s*in	bu*zz*
ʃ (š)	*sh*ine	ma*ch*ine	ca*sh*
ʒ (ž)	—	vi*s*ion	massa*g*e
h	*h*ead	be*h*ind	—
tʃ (č)	*ch*air	tea*ch*er	whi*ch*
dʒ (ǰ)	*j*ump	lar*g*er	hu*g*e
m	*m*ail	re*m*ind	roo*m*
n	*n*est	te*n*or	bea*n*
ŋ	—	a*ng*er	ki*ng*
j (y)	*y*ard	be*y*ond	so*y*
w	*w*ay	re*w*ind	lo*w*
ɹ (r, ɻ)	*r*ain	bo*r*ing	fou*r*
l	*l*ight	bu*ll*et	mai*l*

Vowels

Phonetic symbol	Word positions		
	Initial	**Medial**	**Final**
Vowels and diphthongs			
i (ij, iy)	*ea*se	f*ee*t	b*ee*
ɪ	*i*t	s*i*t	—
e (ej, ei, ey)	*ei*ght	b*a*ke	s*ay*
ɛ	*e*dge	r*e*d	—
æ	*a*nger	n*a*p	—
ʌ	*o*ven	l*o*ve	—
ə	*a*bove	ofte*n*	Tamp*a*
ɑ	*a*rch	f*a*ther	sp*a*
ɔ	*a*ll	h*a*ll	s*aw*
o (ow, oʊ)	*o*at	g*o*at	b*ow*
ʊ	—	b*oo*k	—
u (uw)	*oo*ze	l*oo*se	tw*o*
aɪ (aj, ay)	*i*ce	s*i*de	b*uy*
ɔɪ (ɔj, ɔy, oɪ, oj, oy)	*oi*l	v*oi*ce	b*oy*
aʊ (au, aw)	*ou*t	s*ou*nd	h*ow*

Minimal Pair & Phonemic Contrast

- pin vs. bin --> /pɪn/ vs. /bɪn/
- beat vs. bit --> /bit/ vs. /bɪt/
- budge vs. buzz --> /bʌdʒ/ vs. /bʌz/
- /p/ vs. /b/ /i/ vs. /ɪ/ /dʒ/ vs. /z/

Phoneme vs. Allophone

/p/

[p^h]

pin [p^hɪn]

aspirated

[p]

spin [spɪn]

unaspirated

Complementary Distribution

/p/ → [pʰ]

stressed syllable onset

(not after /s/)

pin [pʰɪn]

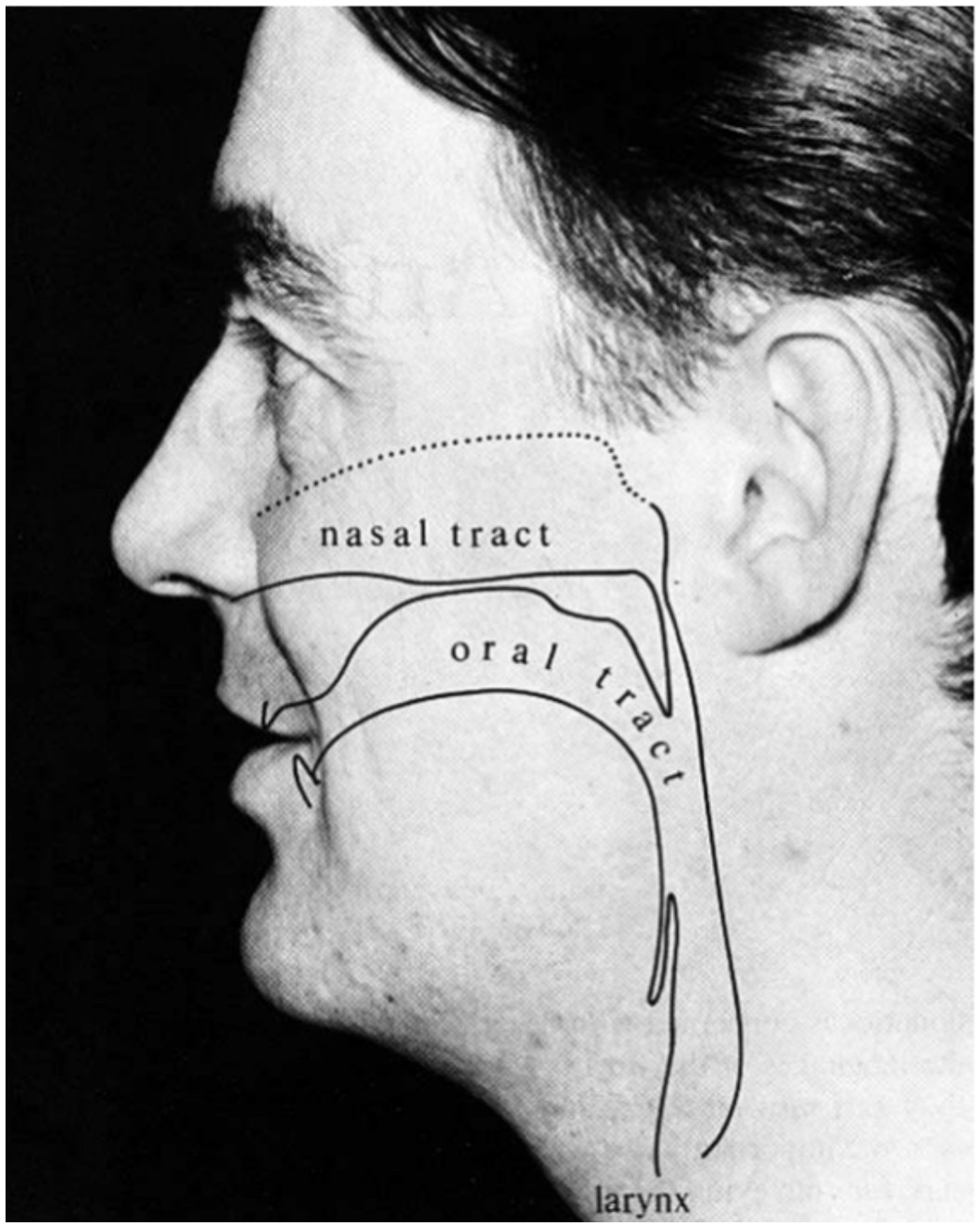
/p/ → [p]

after /s/ in the same onset

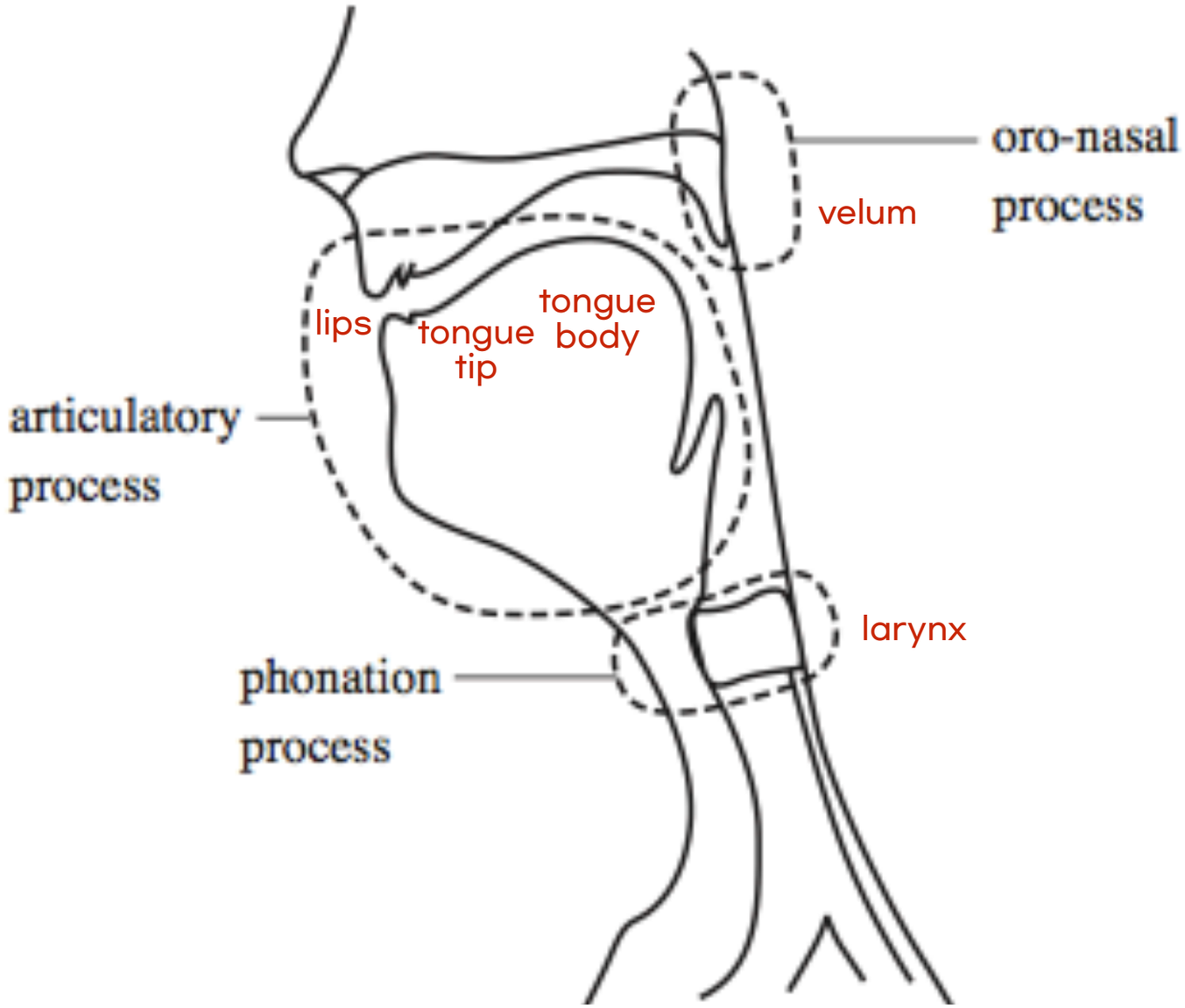
spin [spɪn]

Speech Production

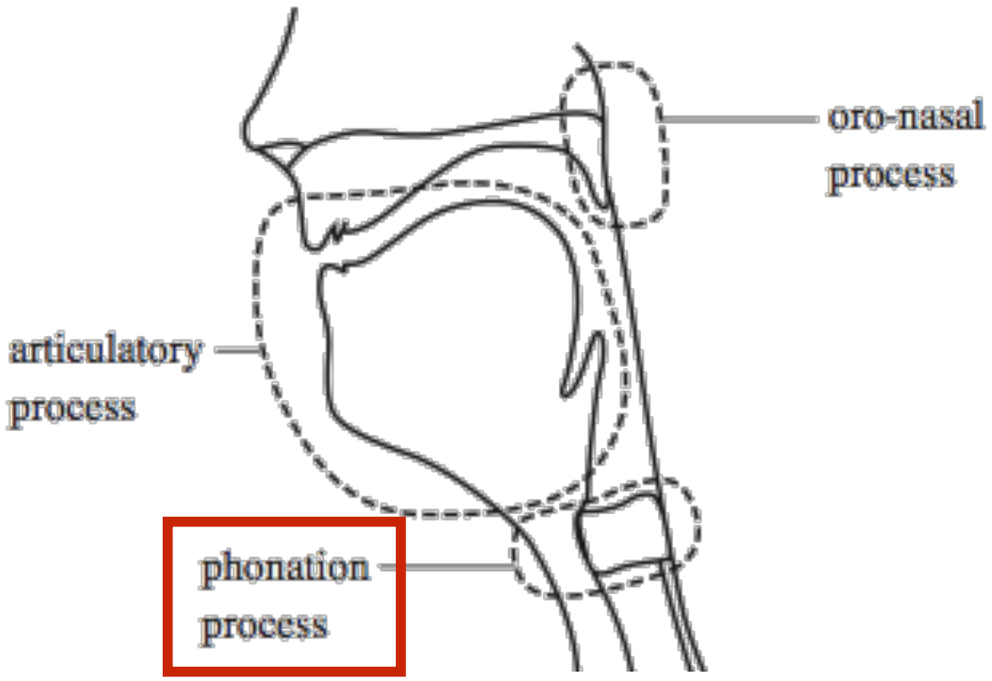
Vocal Tract



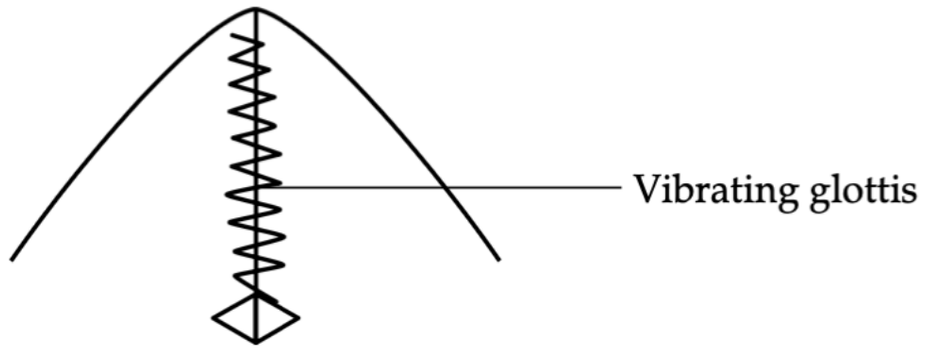
Speech Organs (= Constrictors)



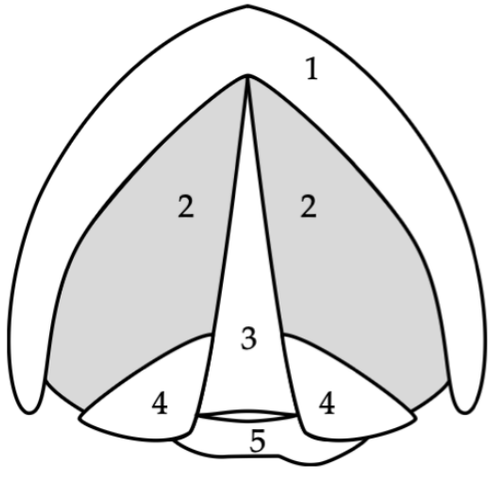
Phonation Process



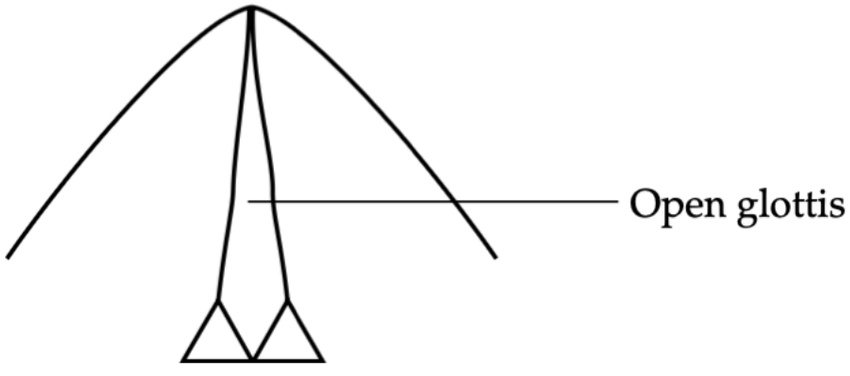
voiced (b, d, g, v, z, a, i, ...)



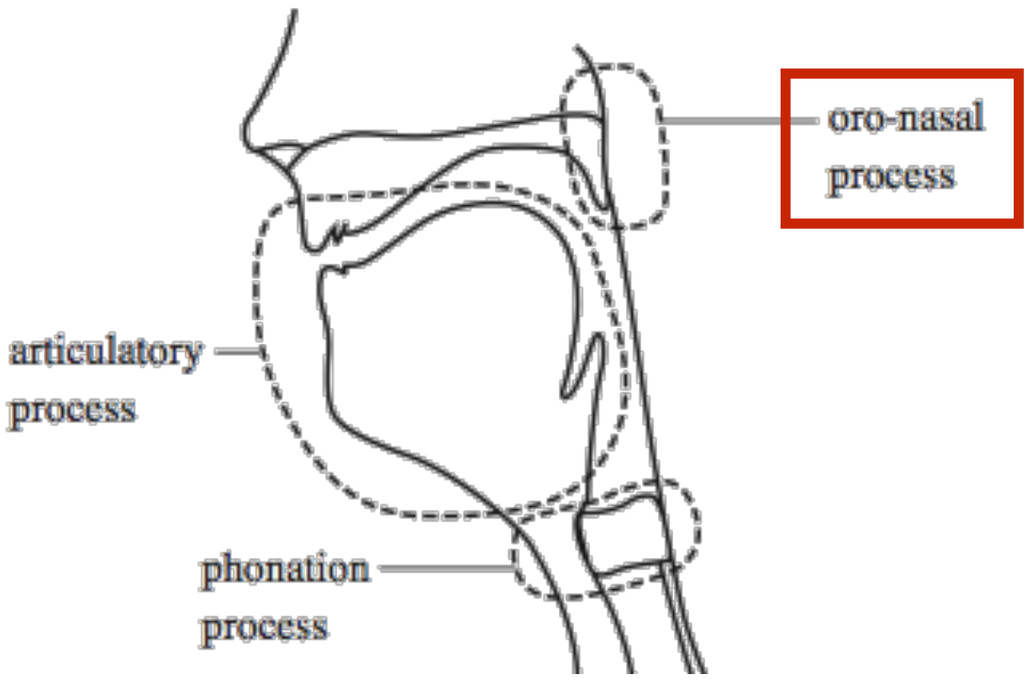
voiceless (p, t, k, f, s, h, ...)



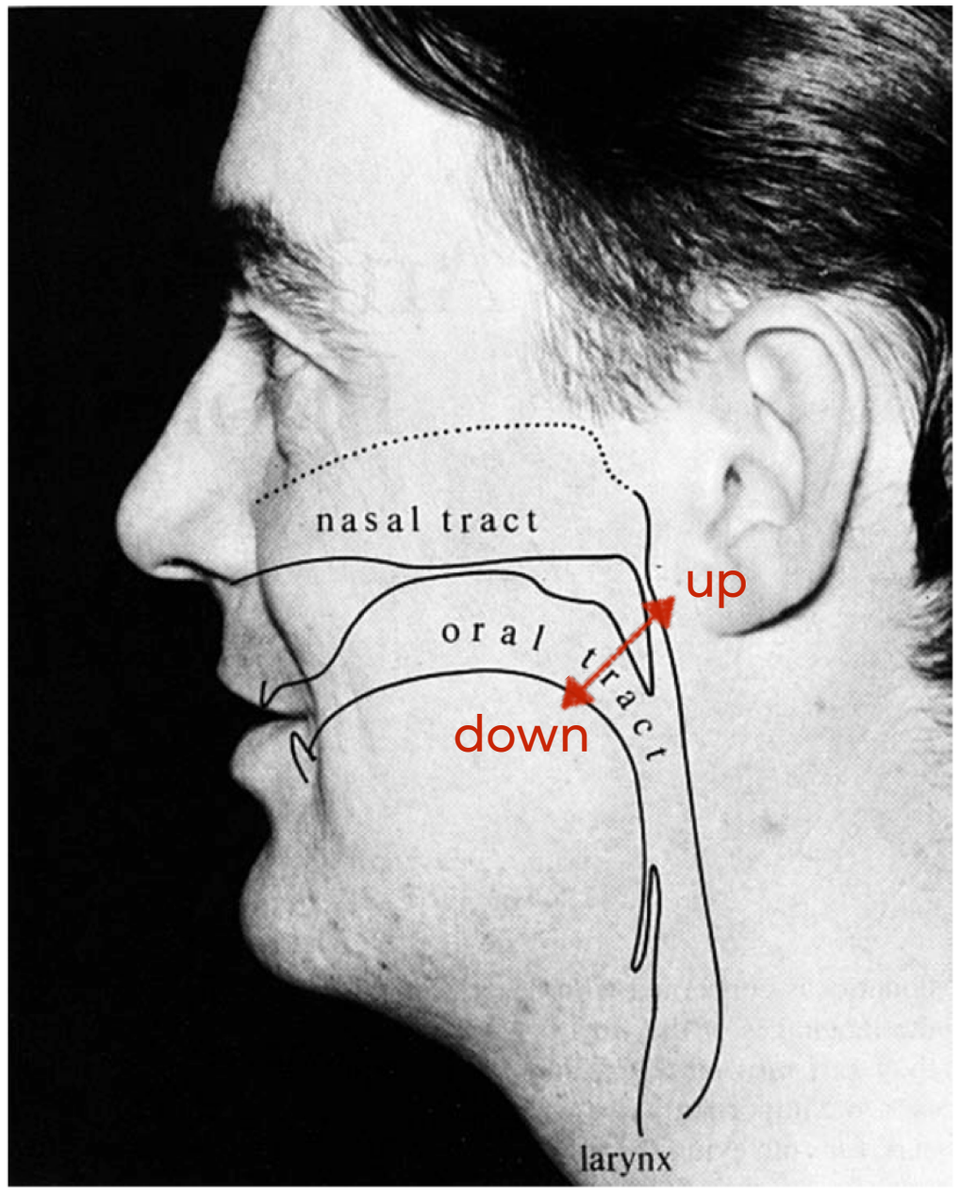
1 Thyroid cartilage
2 Vocal cords
3 Glottis
4 Arytenoid cartilages
5 Cricoid cartilage



Oro-nasal Process



nasals (m, n, ŋ, ...)



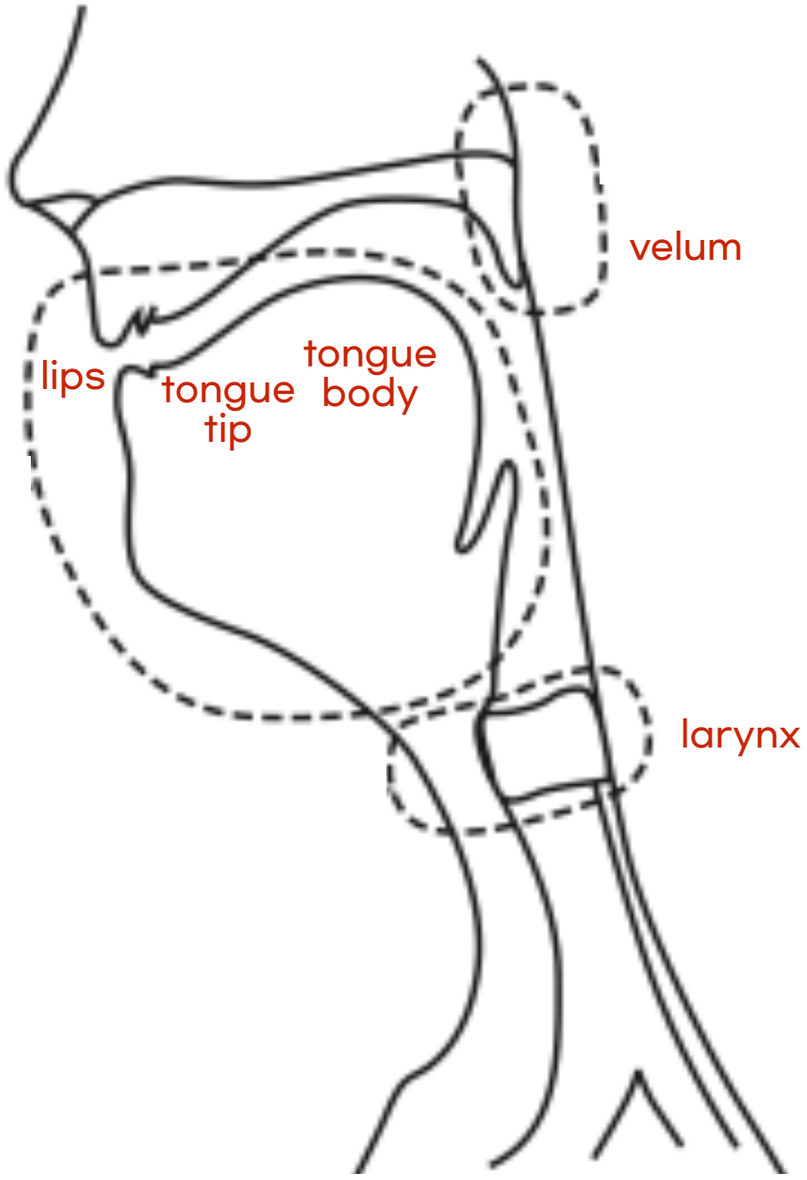
Articulatory Process

Manner of articulation \ Place of articulation	bilabial	labio-dental	dental	alveolar	palato-alveolar	palatal	velar	glottal
nasal (stop)	m			n			ŋ	
stop	p b			t d			k g	ʔ
fricative		f v	θ ð	s z	ʃ ʒ			h
(central) approximant	(w)			r		j	w	
lateral (approximant)				l				

Articulatory Process

a combination of speech organs' actions

- lips (p, b, m, f, v, w, ...)
- tongue tip (t, d, n, r, θ, ð, s, z, ʃ, ʒ, l, ...)
- tongue body (k, g, ŋ, j, vowels, ...)
- velum (nasals: m, n, ŋ, ...)
- larynx (voiced / voiceless)



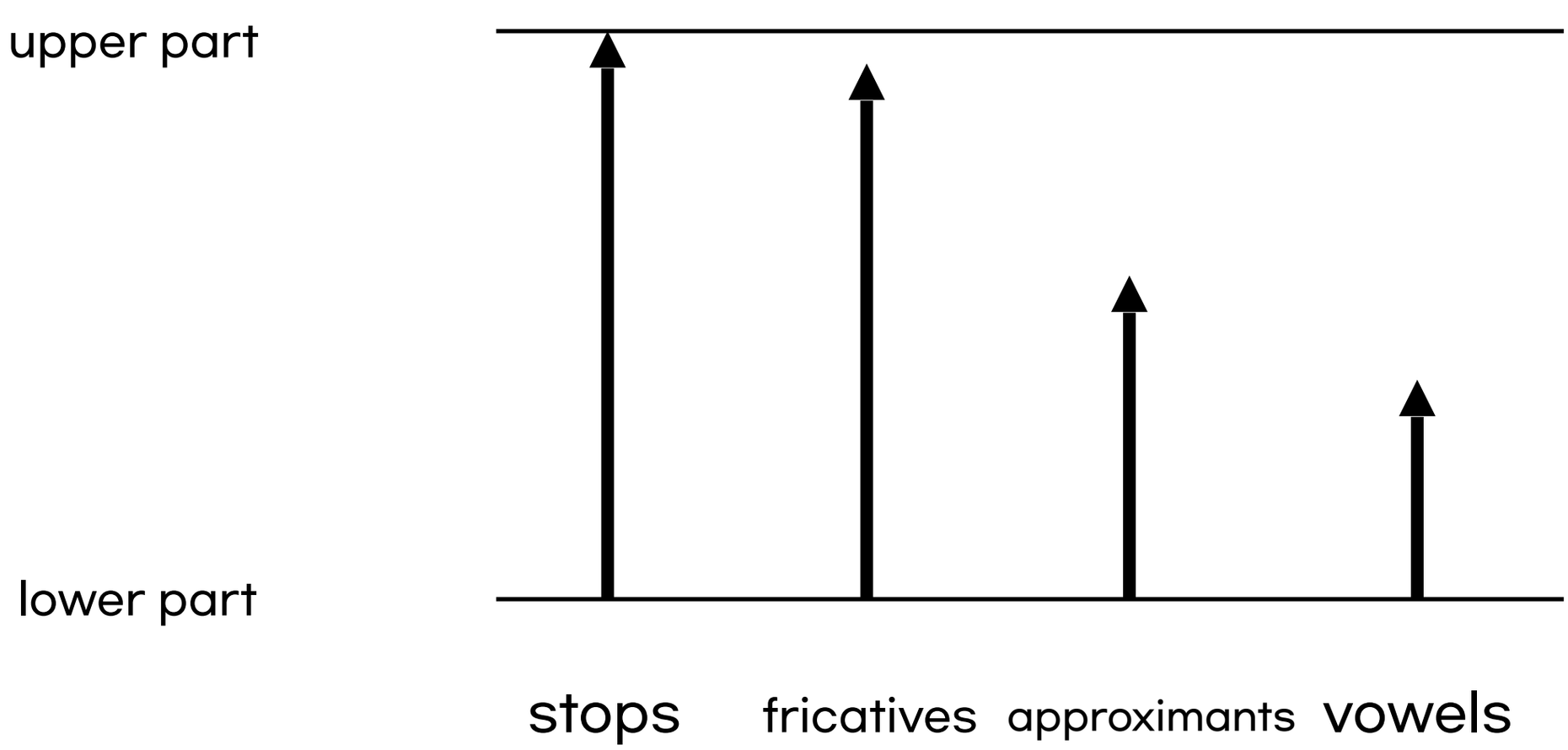
Articulatory Process

- /t/

speech organ	on/off
lips	off
tongue tip	on
tongue body	off
velum	off
larynx	off

- /d/?

Constriction degree (CD)



Constriction Degree (CD)

- stops: full constriction (p, t, k, b, d, g, m, n, ŋ, ⋯)
- fricatives: critical constriction (s, z, f, ʃ, ʒ, θ, ð, ⋯)
- approximants: little constriction (r, l, w, j, ⋯)
- vowels: no constriction (vowels)

/t/ vs. /s/

speech organ	on/off	CD
lips	off	-
tongue tip	on	stop
tongue body	off	-
velum	off	-
larynx	off	-

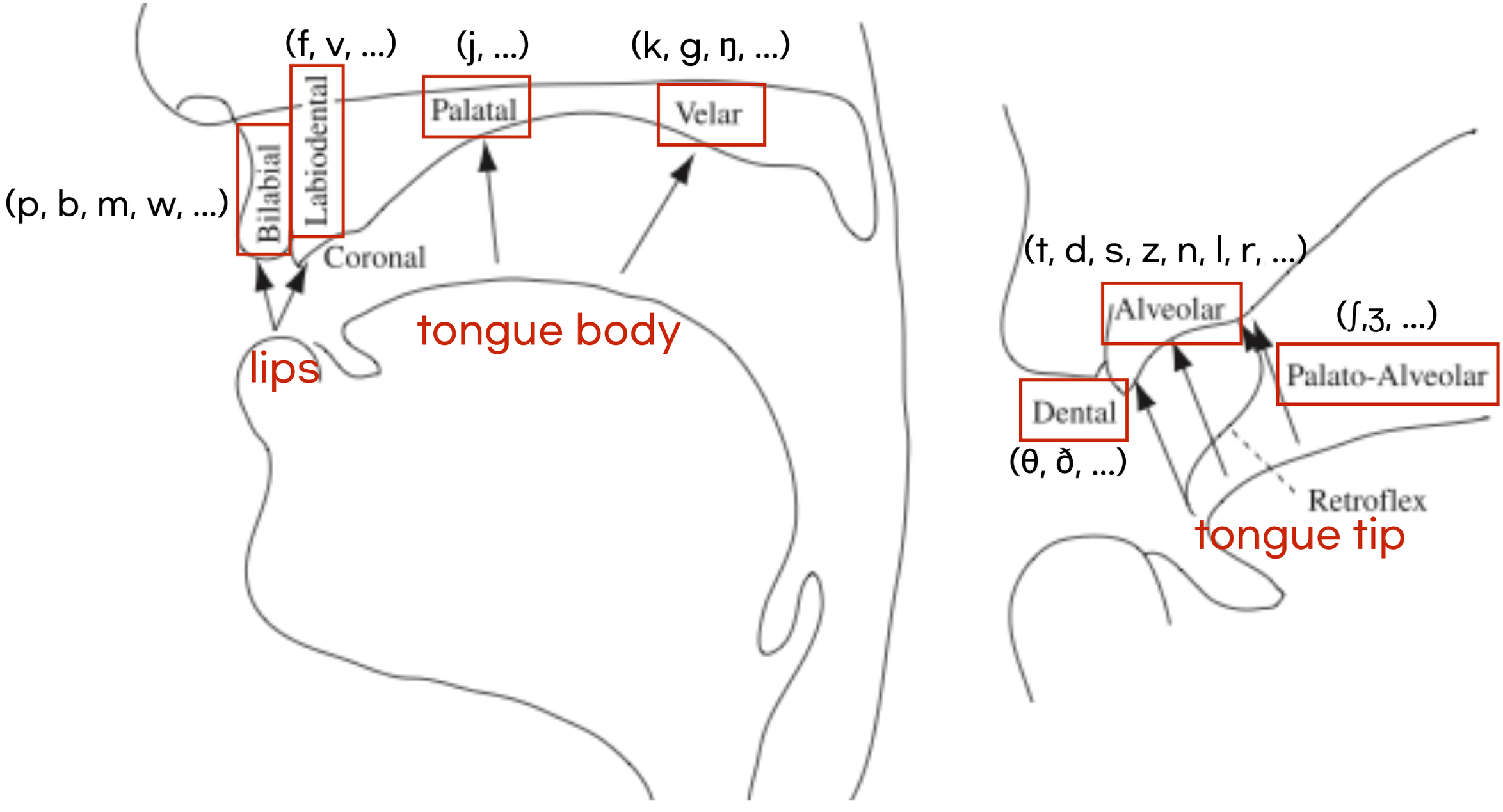
speech organ	on/off	CD
lips	off	-
tongue tip	on	fric.
tongue body	off	-
velum	off	-
larynx	off	-

/a/ vs. /g/

speech organ	on/off	CD
lips	off	-
tongue tip	off	-
tongue body	on	vowel
velum	off	-
larynx	on	-

speech organ	on/off	CD
lips	off	-
tongue tip	off	-
tongue body	on	stop
velum	off	-
larynx	on	-

Constriction Location (CL)



/s/ vs. /ʃ/

speech organ	on/off	CD	CL
lips	off	-	-
tongue tip	on	fric.	alv.
tongue body	off	-	-
velum	off	-	-
larynx	off	-	-

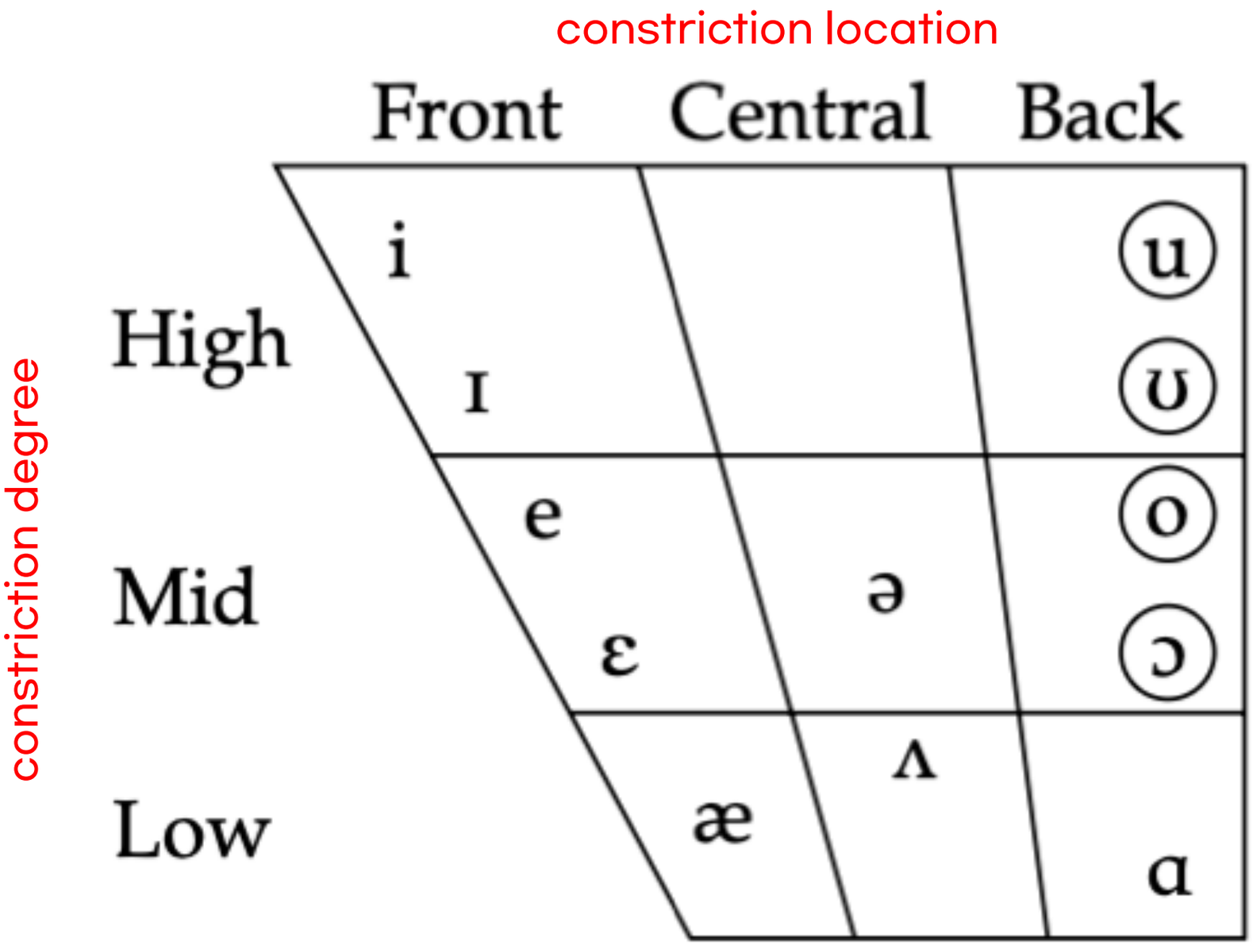
speech organ	on/off	CD	CL
lips	off	-	-
tongue tip	on	fric.	pal.-alv.
tongue body	off	-	-
velum	off	-	-
larynx	off	-	-

Vowels

Figure 1.12 The positions of the vocal organs for the vowels in the words 1 *heed*, 2 *hid*, 3 *head*, 4 *had*, 5 *father*, 6 *good*, 7 *food*. The lip positions for vowels 2, 3, and 4 are between those shown for 1 and 5. The lip position for vowel 6 is between those shown for 1 and 7.



Vowel chart


constriction location
Front
Central
Back
constriction degree
High
Mid
Low
i
u
ɪ
ʊ
e
o
ə
ɛ
ɔ
ʌ
æ
ɑ

Reference

Ladefoged, P., & Johnson, K. (2011). A course in phonetics (6th ed.). Boston, MA: Wadsworth.

Yavaş, M. (2011). *Applied English phonology* (2nd ed.). Malden, MA: Wiley-Blackwell.